Mini-Amazon Milestone 2 Report

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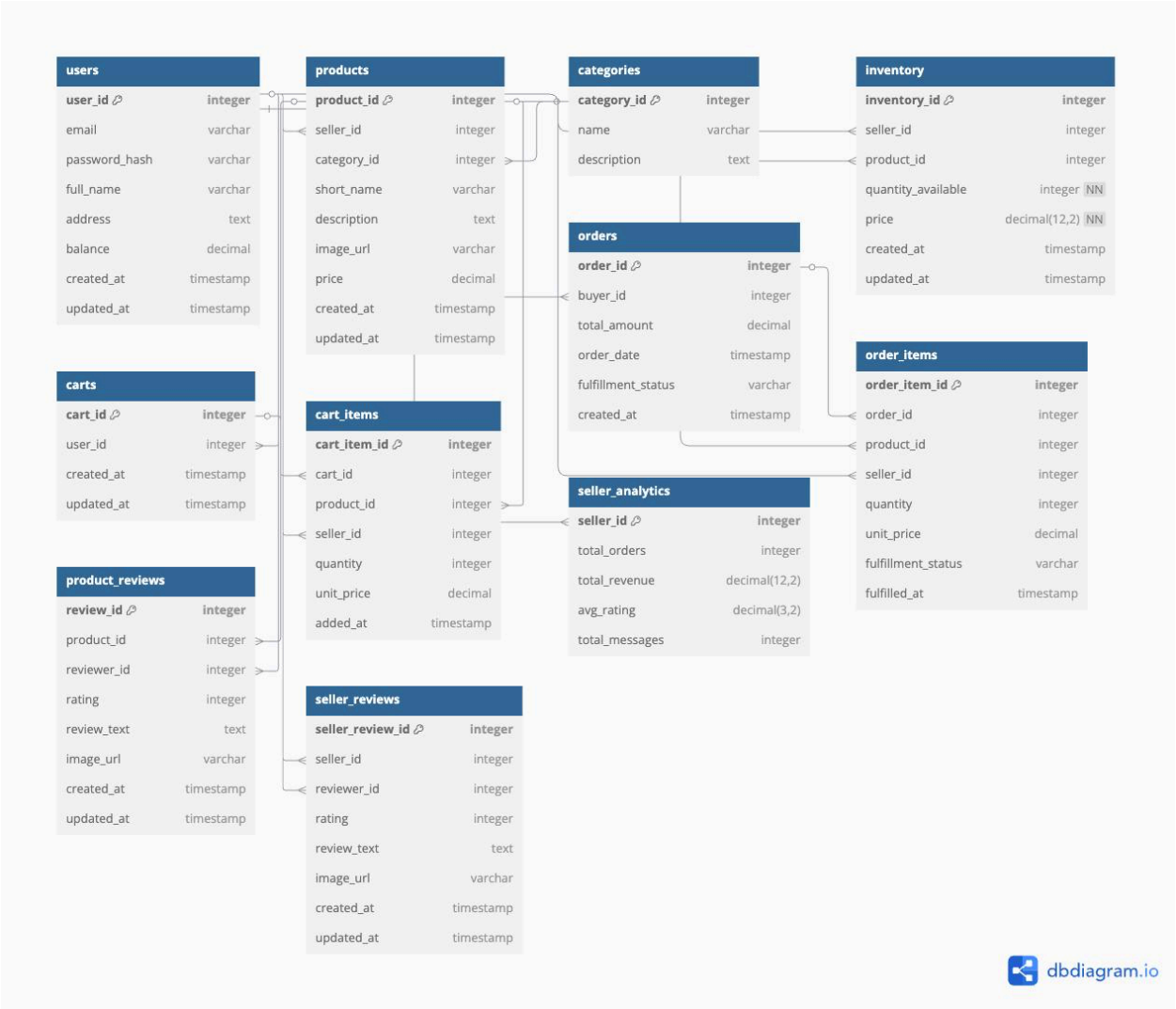
1. Database Schema Design



Users (accounts)

- **user_id** (PK)
- email (unique)
- password_hash
- full_name
- address
- balance (decimal, default 0.00)

- created_at
- Updated_at

Products

- **product_id** (PK)
- seller_id (FK → Users.user_id)
- category_id (FK → Categories.category_id)
- short_name
- description
- image_url
- price (decimal)

Categories

- **category_id** (PK)
- name
- Description

Carts

- **cart_id** (PK)
- user_id (FK → Users.user_id)
- created_at
- updated_at

Cart_Items

- **cart_item_id** (PK)
- cart_id (FK → Carts.cart_id`)
- product_id (FK → Products.product_id`)
- seller_id (FK → Users.user_id`) *(seller of the specific product)*
- quantity (integer)
- unit_price (decimal at the time of addition to cart)
- added_at

Orders

- **order_id** (PK)
- buyer_id (FK → Users.user_id`)
- total_amount (decimal)
- order_date
- fulfillment_status (pending, partial, fulfilled)
- created_at

Order_Items

- **order_item_id** (PK)
- order_id (FK → Orders.order_id`)

- product_id (FK → Products.product_id`)
- seller_id (FK → Users.user_id`)
- quantity (integer)
- unit_price (decimal, finalized at time of purchase)
- fulfillment_status (pending, fulfilled)
- fulfilled_at (timestamp, nullable)

Product_Reviews

- **review_id** (PK)
- product_id (FK → Products.product_id`)
- reviewer_id (FK → Users.user_id`)
- rating (integer, e.g., 1–5)
- review_text
- image_url (optional, nullable)
- created_at
- updated_at

(Unique constraint: reviewer_id + product_id)

Seller_Reviews

- **seller_review_id** (PK)
- seller_id (FK → Users.user_id`)
- reviewer_id (FK → Users.user_id`)
- rating (integer, e.g., 1–5)
- review_text
- image_url (optional, nullable)
- created_at
- updated_at

(Unique constraint: reviewer_id + seller_id)

Inventory Table

- id (PK, auto-incrementing)
- seller_id (FK → Sellers.id, NOT NULL)
- product_id (FK → Products.id, NOT NULL)
- quantity_available (integer, NOT NULL, CHECK quantity_available >= 0)
- price (DECIMAL(12,2), NOT NULL, CHECK price >= 0)
- created_at (timestamp, DEFAULT CURRENT_TIMESTAMP)
- updated_at (timestamp, DEFAULT CURRENT_TIMESTAMP on update)
- (Unique constraint: seller_id + product_id)

Seller_Analytics Table (Optional)

- seller_id (PK, FK → Sellers.id)
- total_orders (integer, DEFAULT 0, CHECK total_orders >= 0)
- total_revenue (DECIMAL(12,2), DEFAULT 0, CHECK total_revenue >= 0)
- avg_rating (DECIMAL(3,2), CHECK avg_rating BETWEEN 0 AND 5)
- total_messages (integer, DEFAULT 0, CHECK total_messages >= 0)

Carts

- **cart_id** (PK)
- user_id (FK → Users.user_id)
- created_at
- updated_at

Cart_Items

- **cart_item_id** (PK)
- cart_id (FK → Carts.cart_id`)
- product_id (FK → Products.product_id`)
- seller_id (FK → Users.user_id`) *(seller of the specific product)*
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(Unique constraint: reviewer_id + product_id)

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2. Website Design

https://www.figma.com/design/NzYEK7wwVlGKucxSJNaEU3/Untitled?node-id=0-1&t=X5rMqvtdiPGLgiNI-1